

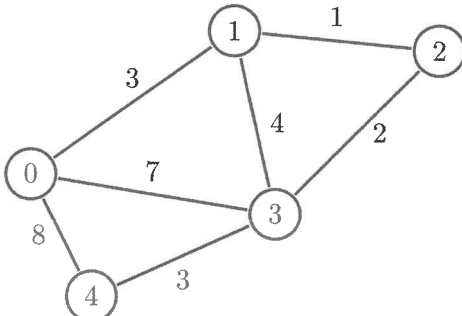
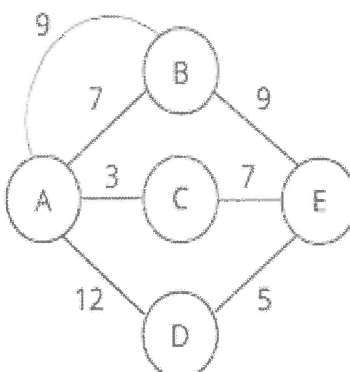
**RAMAIAH**

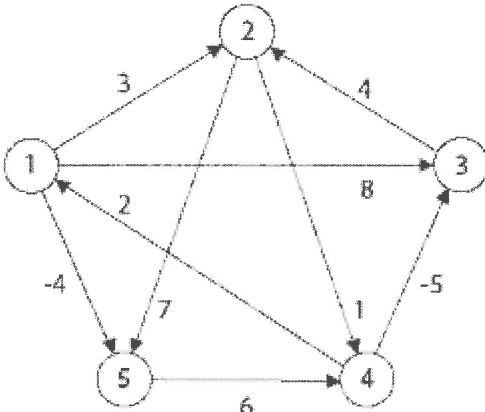
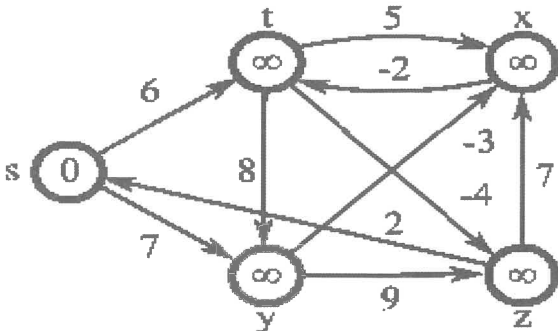
Institute of Technology

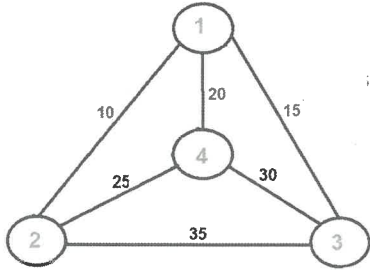
Department of Computer Science & Engineering**QUESTION BANK FOR IV SEMESTER****Subject Code: 24CSL46****TERM: Feb – June 2026****I.A. Marks: 50****Subject Name: Design and Analysis of Algorithms Laboratory****Exam Hours: 03****Credits: 0:0:1****Exam Marks: 50****Note:**

- 1) One Question to be selected from the lot. Algorithm for the same should be written.
- 2) Programs should be written or implemented using JAVA Programming Language.

SI No.	List of Programs																																																		
1.	A logistics company warehouse stores parcel weights in an unsorted order in its database. To organize the parcels efficiently for shipment, the system selects one parcel weight as a pivot and partitions the list by scanning from both ends toward the center, placing lighter parcels on the left side and heavier parcels on the right side. This partitioning process continues recursively until the entire list of parcel weights becomes sorted. Use the Quicksort algorithm with Hoare’s partition scheme to solve the problem. Implement the algorithm using Java to sort the parcel weights and display the execution time taken by the program. Input Data (Sample) Parcel weights in kilograms: [42, 15, 67, 23, 89, 4, 55, 31, 72, 18]																																																		
2.	Given a set of men and women design and implement Gale–Shapley algorithm to determine the stable set of marriages among them. Assumptions: Men propose first according to their preference list. Women can choose a better partner based on the preference. Use Stable Matching Gale Shapely algorithm to solve the given problem. Implement the algorithm using java program to match the preferences perfect and stable. Men’s preference list <table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>A</td><td>W</td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>B</td><td>X</td><td>W</td><td>Y</td><td>Z</td></tr><tr><td>C</td><td>Z</td><td>W</td><td>Y</td><td>X</td></tr><tr><td>D</td><td>Z</td><td>Y</td><td>X</td><td>W</td></tr></table> Women’s preference list <table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>W</td><td>A</td><td>B</td><td>D</td><td>C</td></tr><tr><td>X</td><td>B</td><td>C</td><td>A</td><td>D</td></tr><tr><td>Y</td><td>C</td><td>D</td><td>A</td><td>B</td></tr><tr><td>Z</td><td>C</td><td>D</td><td>B</td><td>A</td></tr></table>		1	2	3	4	A	W	X	Y	Z	B	X	W	Y	Z	C	Z	W	Y	X	D	Z	Y	X	W		1	2	3	4	W	A	B	D	C	X	B	C	A	D	Y	C	D	A	B	Z	C	D	B	A
	1	2	3	4																																															
A	W	X	Y	Z																																															
B	X	W	Y	Z																																															
C	Z	W	Y	X																																															
D	Z	Y	X	W																																															
	1	2	3	4																																															
W	A	B	D	C																																															
X	B	C	A	D																																															
Y	C	D	A	B																																															
Z	C	D	B	A																																															
3.	A university examination cell must schedule several evaluation activities such that some activities are completed before others begin. These activities and their precedence relations are represented using a directed graph, where each vertex represents an activity and each directed edge represents the order in which activities must be performed.																																																		

	Consider the following directed graph: Activities: - A – Collect answer scripts - B – Verify attendance sheets - C – Allot scripts to evaluators - D – Begin valuation - E – Moderation process - F – Final result preparation Dependencies: - A → C - B → C - C → D - D → E - E → F Use Topological Sorting with DFS algorithm to determine a valid order in which the activities can be performed. Implement the algorithm using Java and analyse the time and space complexity.
4.	A truck driver is given a set of locations with distances between them. The company strictly orders that the truck must start from a specific source location and reach all other locations using the minimum possible distance. .Use Dijkstra’s algorithm to find the shortest distance from the source (Source = 0) to all other locations and print the intermediate locations in the path. Implement the algorithm using java and display the execution time taken by the program.  <pre> graph LR 0((0)) --- 3 1((1)) 0 --- 7 3((3)) 0 --- 8 4((4)) 1 --- 1 2((2)) 1 --- 4 3 2 --- 2 3 3 --- 3 4 </pre>
5.	A renewable energy company plans to connect all its solar power stations spread across different locations using transmission lines. The cost of laying transmission lines between every pair of stations is known. The company wants to ensure that all stations are connected with minimum total installation cost, and the network construction can begin from any station. Use Prim’s algorithm to form the minimum-cost transmission network. Implement the algorithm using Java and display the running time of the algorithm.  <pre> graph TD A((A)) --- 7 B((B)) A --- 3 C((C)) A --- 12 D((D)) B --- 9 E((E)) C --- 7 E C --- 5 D D --- 5 E A -.- 9 B </pre>

6.	A plagiarism detection system searches for a particular word Pattern = "ENGINE" in large documents Text = "COMPUTER SCIENCE AND ENGINEERING". The system compares characters from right to left and skips multiple characters whenever a mismatch occurs in order to speed up the search process. Use Horspool's String Matching Algorithm to solve the given problem. Implement the algorithm using Java to efficiently perform the string matching operation and determine whether the pattern exists in the text. Also evaluate the time and space complexity of the algorithm.															
7.	A network administrator needs to determine the shortest communication path between every pair of servers in a network. Apply the Dynamic Programming technique using the Floyd–Warshall algorithm to solve the All-Pairs Shortest Path problem. Implement the algorithm using Java to compute the shortest paths between all pairs of servers, and analyze its time and space complexity. 															
8.	A trekking enthusiast carries a backpack with a maximum capacity of 7 kg, where each item has a specific weight and value. Apply the Dynamic Programming technique using Memory Functions to solve the Knapsack problem while avoiding repeated subproblem computation. Implement the algorithm using Java to determine the most valuable set of items without exceeding the backpack capacity, and analyze its time and space complexity. <table><tr><th>Item</th><th>Weight</th><th>Value</th></tr><tr><td>1</td><td>3</td><td>10</td></tr><tr><td>2</td><td>5</td><td>4</td></tr><tr><td>3</td><td>6</td><td>9</td></tr><tr><td>4</td><td>2</td><td>11</td></tr></table>	Item	Weight	Value	1	3	10	2	5	4	3	6	9	4	2	11
Item	Weight	Value														
1	3	10														
2	5	4														
3	6	9														
4	2	11														
9.	A financial transaction network consists of multiple routes, some of which have negative transaction costs. Apply the Dynamic Programming technique to solve the Bellman–Ford shortest path problem. Implement the algorithm using Java to compute the shortest transaction cost from a given source and detect negative-cost cycles, and analyze its time and space complexity. 															

10.	A chess puzzle requires placing queens on an $N \times N$ chessboard such that no two queens attack each other. Apply the Backtracking technique to solve the N-Queens problem ($N=4$). Implement the algorithm using Java to display all valid queen arrangements, and analyze its time and space complexity.
11.	A financial auditing system needs to determine whether a subset of transaction values sums exactly to a suspicious target amount. Apply the Backtracking technique to solve the Subset Sum problem. Implement the algorithm using Java to identify whether such a subset exists, and analyze its time and space complexity. Sample Set = {10, 7, 5, 18, 12, 20, 15}, Target Sum = 35
12.	A courier company aims to determine the shortest possible delivery route for its delivery agent. The agent must visit each city exactly once and return to the starting city while eliminating routes whose costs exceed the current minimum cost. Apply the Branch and Bound technique to solve this Travelling Salesman Problem (TSP). Implement the algorithm using Java to determine the shortest route and the minimum delivery cost. 

Marks Distribution:

Write-Up	Execution	Viva	Change of Program	Total
8 Marks	35 Marks	7 Marks	-8 Marks	50 Marks

Prepared by: Dr. S Rajarajeshwari

Narayan
HOD CSE 02/06/26.

Reviewed By:

J. P.
02/06/26